Math for CS & AI: Homework 6

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(Adapted from earlier versions of this course taught by Andrew Yao)

Due Date: Apr 12, 2026

Optional Reading Task

This reading task is optional, but if the $\sum$ and $\prod$ symbols look a bit scary to you, we highly recommend going through this material first before working on generating function problems.

- Read Chapter 2 of the following book (available on the course website):

  Concrete Mathematics: A Foundation for Computer Science, by Ronald Graham, Donald Knuth, and Oren Patashnik

Problem Set

- Exercises 3.12–3.14 from Chapter 3 and Exercises 4.1 from Chapter 4 of the lecture notes:

  https://cloud.tsinghua.edu.cn/d/57ddec714cf74c1cab85/

Writing Guidelines

- Same as the first homework.
- We may deduct up to 3 points for poor writing.
- **Special Note:** Your submission should contain only a single PDF file. Please do not submit other files such as `*.zip` or `*.tex`.

Additional Hints (Experimental)

- We provide additional hints for some of the homework problems in the next few pages.
- If you get stuck on a problem, scroll down to find that problem and read the hints.

- Scroll *very slowly* so that you do not accidentally see hints for problems you have not thought about yet.
- Homework is meant to help you practice your problem-solving skills. It is your job to make sure you have thought enough before reading the hints. Ultimately, there will be no additional hints and no AI tools in the exams.
- Again, **peer learning is strongly encouraged**. Reading hints or getting help from AIs is not the same as learning from your classmate who has thought about the problem and can explain it to you. Make some friends who are willing to discuss problems or even have hotpot with you. Learn from their way of thinking, including both successful and unsuccessful attempts.

The rest of this document contains hints for the homework problems.



Think twice before scrolling down!

Hints

Problem 3.12

- $H(p) \geq 0$ is trivial, right? Just check $p(x) \log_2 \frac{1}{p(x)} \geq 0$.
- For proving $H(p) \leq \log_2 |\mathcal{X}|$, apply Jensen's inequality to $\mathbb{E}[\log_2 \frac{1}{p(X)}]$.
- For proving AEP, take logarithm and use Chernoff bounds / Hoeffding's inequality.

Problem 3.13

- Just encode the strings in the typical set. How many are they?
- For the strings that are not in the typical set, encode them in the most naïve way.

Problem 3.14

- Do not overthink. Just use the definition of entropy.

$$H(p_Y) = \sum_y p_Y(y) \log_2 \frac{1}{p_Y(y)} = \sum_x p_X(x) \log_2 \frac{1}{p_Y(f(x))} \leq \ ?$$

- The entropy of the input and output strings are the same.
- For the coding scheme, first argue that the maximum length is at most $\lceil n \log_2 |\mathcal{X}| \rceil$. Otherwise we can always find a coding scheme with a shorter expected length.
- If the expected length is L, then the entropy can be bounded by L plus the entropy of the string length. The latter can be bounded by $\log$ (# of possible string lengths).

Problem 4.1

- Let $c_n = a_{2n}$. Compute the generating function of c_n first.
- Two cases: the last two columns are formed by 3 horizontal dominoes, or the last $2k$ columns are formed by 2 vertical dominoes and many horizontal dominoes.
- Write a recurrence for c_n. It may involve a prefix sum.
- If you see some $\sqrt{3}$ in your final answer, then you are probably right.